

Autoantigens and Immune Responses in Adamantiades-Behçet Disease

Tropomyosins and the 160-kd polypeptide kinectin have been identified as autoantigens in Adamantiades-Behçet disease.

Heat Shock Proteins

Elevated levels of heat shock protein (HSP)-specific antibodies are found in the serum of patients with Adamantiades-Behçet disease. T cells react to the 60-kd HSP, and T-cell epitope mapping has revealed four distinct peptide determinants within this protein that may contribute to disease pathogenesis.

Cytokine Mediators

Patients exhibit increased serum levels of pro-inflammatory cytokines, including IL-1, IL-8, IL-12, IL-17, and tumor necrosis factor- α . IL-8 is particularly significant due to its potent pro-inflammatory effects, ability to be secreted by endothelial cells, and role as a sensitive marker of disease activity. Cytokine production may vary depending on the organ system involved.

Endothelial Cells

The endothelium appears to be a primary target in Adamantiades-Behçet disease, possibly due to dysregulated immune activity. An IgM-type autoantibody against a 47-kd cell surface HSP associated with endothelial α -enolase has been detected in patient sera. Plasma endothelin-1 levels are significantly elevated, suggesting vasoconstriction and increased synthesis by damaged vascular endothelial cells. Thrombomodulin, a glycoprotein expressed on vascular endothelium, is also increased in plasma during active disease, indicating endothelial injury.

Clinical Findings

Adamantiades-Behçet disease is a chronic, relapsing, multisystem disorder that can be life-threatening. Common manifestations include recurrent oral aphthous ulcers, genital ulcers, skin lesions, ocular inflammation (such as iridocyclitis or posterior uveitis), and arthritis or arthropathy. Additional features may involve the vascular, gastrointestinal, neurologic, psychiatric, pulmonary, renal, and cardiac systems, as well as epididymitis. The full clinical spectrum typically emerges within months of initial symptoms, though presentation can range from acute multisystem involvement to gradual progression over years.

Approach to the Patient with Oral Ulcers, Genital Ulcers, or Iridocyclitis/Posterior Uveitis

Diagnosis of Adamantiades-Behçet disease relies on clinical criteria, as no pathognomonic laboratory or histologic findings exist. The most widely used diagnostic criteria have been those established by the International Study Group and the Behçet Disease Research Committee of Japan. However, limitations in sensitivity and specificity have led to revisions of these criteria.

Revised International Criteria for Behçet's Disease (ICBD)

Diagnosis is established with a total score of 3 points or more:

- **1 point:** Oral aphthosis (recurrent oral ulcers)
- **1 point:** Skin manifestations (e.g., pseudofolliculitis, erythema nodosum-like lesions)
- **1 point:** Vascular lesions (e.g., phlebitis, superficial or deep vein thrombosis, arterial thrombosis, aneurysm)

- **1 point:** Positive pathergy test
- **2 points:** Genital aphthosis (ulcers or scars)
- **2 points:** Ocular lesions (anterior or posterior uveitis, retinal vasculitis)

These revised criteria aim to improve diagnostic accuracy and applicability across diverse populations.